Edge AI Multi-Model Scheduler

Resource Management System · Project Abstract

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PROJECT DESCRIPTION

This project focuses on building an intelligent Edge AI scheduling system that efficiently manages and executes multiple AI models on resource-constrained devices—addressing performance bottlenecks and system instability. The Edge AI Multi-Model Scheduler acts as a centralized orchestration layer that dynamically allocates computational resources (CPU and GPU) to different AI models based on priority, workload, and system conditions. By integrating adaptive scheduling strategies and real-time monitoring, the system ensures optimal performance, reduced latency, and stable execution even under heavy workloads.

PROJECT SCENARIO

Edge devices such as embedded systems or IoT hardware often run multiple AI models simultaneously—for tasks like object detection, analytics, and monitoring. Limited computational resources can lead to performance degradation, overheating, or system crashes when workloads are unmanaged. The scheduler addresses this by intelligently distributing resources among competing models: a high-priority real-time model (e.g., safety detection) receives more GPU allocation, while lower-priority background models run at reduced frequency. The system continuously monitors resource utilization and dynamically adjusts execution parameters such as frame rate and model priority, ensuring smooth operation and consistent performance across all deployed AI workloads.

KEY FUNCTIONS & MODULAR COMPONENTS

Dynamic Scheduling Engine
Allocates resources to multiple AI models based on priority, workload, and system conditions.

Priority-Based Execution
Ensures critical models receive higher computational preference over non-critical tasks.

Resource Monitoring Tool
Tracks real-time CPU and GPU usage to prevent overload and optimize allocation.

Load Balancing Module
Distributes workloads efficiently across processing units to maintain system stability.

Adaptive FPS Control
Dynamically adjusts frame rates to balance performance and resource consumption.

Performance Logging
Records execution metrics and system behavior for analysis and optimization.

TOOLS & TECHNOLOGIES

Language Python (Multiprocessing)	**Inference** ONNX Runtime	**Platform** Jetson Nano (Edge Device)
Monitoring psutil / NVIDIA Tools	**Logging** Logging Frameworks	**Dashboard** Flask / React (Optional)